MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548638

(DELAY, FINE/FINE)

1. .SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the fine/fine delay assembly used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548638	Delay, Fine/Fine

PACKAGING DATA SHEET

10548638	Packaging of Delay, Fine/Fine

(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548638 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - For the requirements of 3.5.2 through 3.5.6.2 logic input and output levels shall be as follows:

Logic 0: 0.0 ±0.5 Vdc level
Logic 1: +3.5 ±1.0 Vdc level

3.5.1 Input power. - The input power applied between terminals P1-30, P1-31, and P1-32 (+5 V) and terminals P1-24, P1-25 and P1-26 (+5 V return) shall be 5.0 ±0.3 volts direct current (Vdc), at 200 milliamperes (mA) regulated; with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signals. - Input signals to the fine/fine delay assembly shall be: $\overline{T1}$, T11, F1/5, $\overline{F1/5}$, F2/5, $\overline{F2/5}$, and $\overline{F4/5}$ as shown in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ±0.5 volt (V) to 3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in direct current (dc) level, from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V. For the requirements of 3.5.2 through 3.5.5 signal T1, as shown in Figure 1, and a logic 1 shall be applied to P1-34 and P1-19 respectively. For the requirements of 3.5.6 through 3.5.6.2, signal T11, as shown in Figure 1, and a logic 0 shall be applied to P1-19 and P1-34 respectively.

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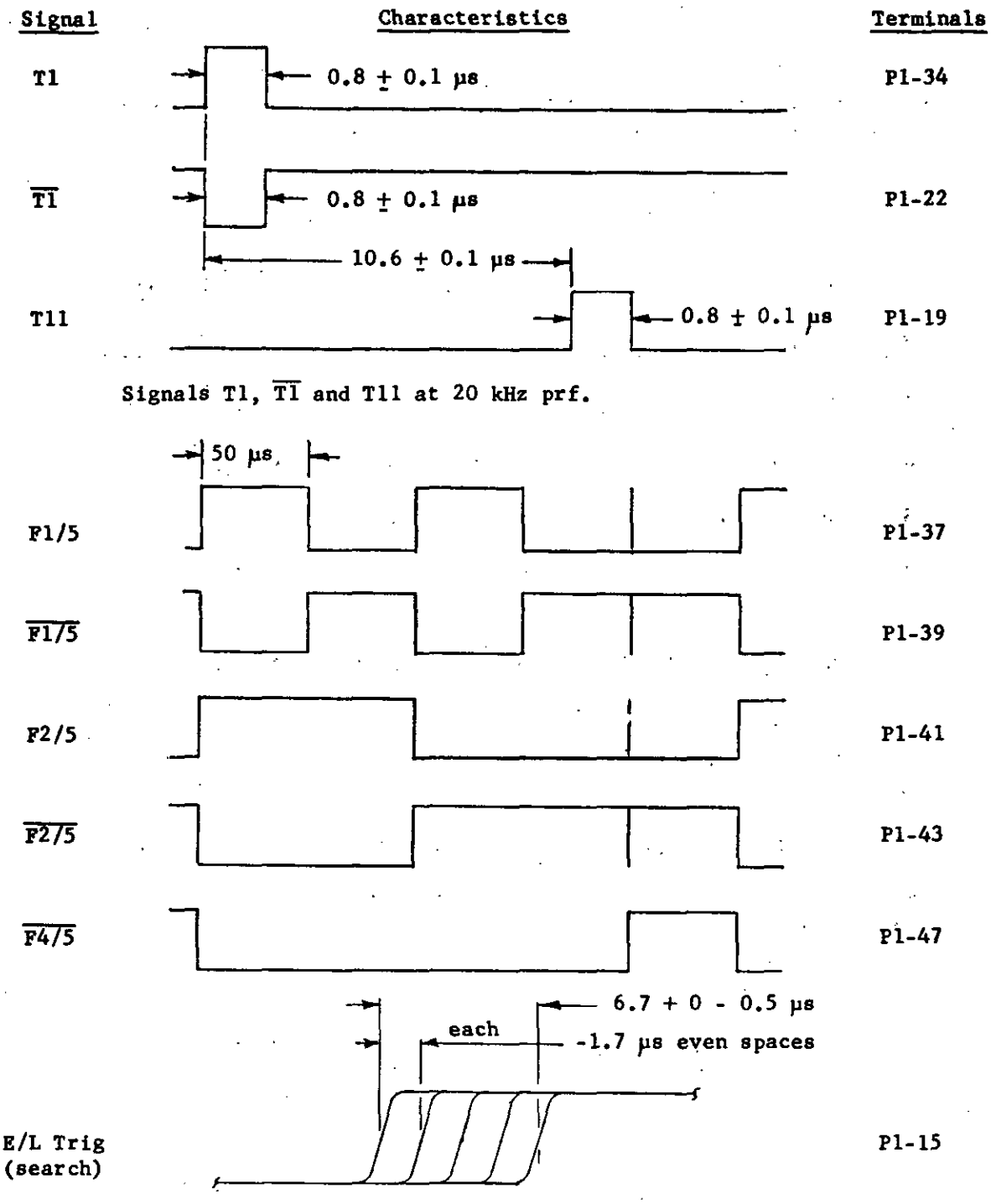


FIGURE 1. INPUT AND OUTPUT SIGNALS.

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3.5.3 RCFD decoder output. - With the input logic levels applied sequentially, as specified in Table I, the range control fine delay (RCFD) decoder output signals shall be as specified in Table I.

TABLE I. - RCFD DECODER OUTPUT SIGNALS

	INPUT TERMINALS								SIGNAL ON OUTPUT TERMINALS							
STEP	42	51	56	44	49	46	54	52	5	6	7	8	9	10	11	12
1	0	0	0	1	1	1	1	0	0	1	1	1	0	1	1	1
2	1	0	1	1	0	1	0	0	1	0	1	1	1	0	1	1
3	0	1	0	0	1	0	1	1	1	1	0	1	1	1	0	1
4	1	1	1	0	0	0	0	1	1	1	1	0	1	1	1	0

3.5.4 T_o 2.0 μs delay output. - The delay of the positive going slope of the negative pulse at P1-36 shall be adjustable by R11 to 2.0 ±0.1 microseconds (μs) (50 percent points) from the leading edge of the input signal at P1-34. The output pulse shall be at the same 20 kilohertz (kHz) pulse repetition frequency (PRF) as the input signal.

3.5.5 T_o 0.2 μs delay output. - At P1-35 a negative pulse shall be present. It shall have an 0.2 ±0.1 μs duration and the negative going slope shall be coincident with the positive going slope of the output signal at P1-36.

3.5.6 R/L trigger output.

3.5.6.1 Fine/fine delay steps. - With a logic 0 applied to P1-3, the E/L trigger output signal at P1-15 shall consist of five leading edges as shown in Figure 1. These leading edges shall be adjustable by C1 through C5 to the limits shown in Figure 1.

3.5.6.2 E/L trigger pulse parameters. - The E/L trigger output signal at P1-15 shall be 0.45 ±0.15 μs wide at a 20 kHz PRF. With a logic 1 applied to P1-33 there shall be a delay of 15 to 35 μs between the leading edge of the input signal at P1-19 and the leading edge of the E/L output signal at P1-15.

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With a logic 0 applied to P1-3, there shall be an additional delay of 15 to 120 μs from the delay present with logic 1 applied to P1-3.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-56652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

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4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table II shall be utilized in the performance of the applicable tests as specified in 4.6.3.

TABLE II. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity, 200 mA.
2. Dual pulse generator	Two separate output pulses, 20 kHz PRF, independently variable in delay to 10.6 μs ±0.1 μs, duration 0.8 ±0.1 μs, with amplitudes of +3.5 ±1.0 Vdc into 50-ohm load.
3. Oscilloscope	Dual channel; rise time of scope plus measuring probe, 10 ns or less; sweep period, 50 ns/cm or greater; input resistance, 8 megohms or greater.
4. Sampling oscilloscope	Rise time of scope plus measuring probe 100 ps or less, sweep period 1 ns/cm or greater; input resistance greater than 500 ohms, input capacitance less than 10 pF.
5. Test fixture	As specified in Figure 2; see Table III for test fixture parts list.

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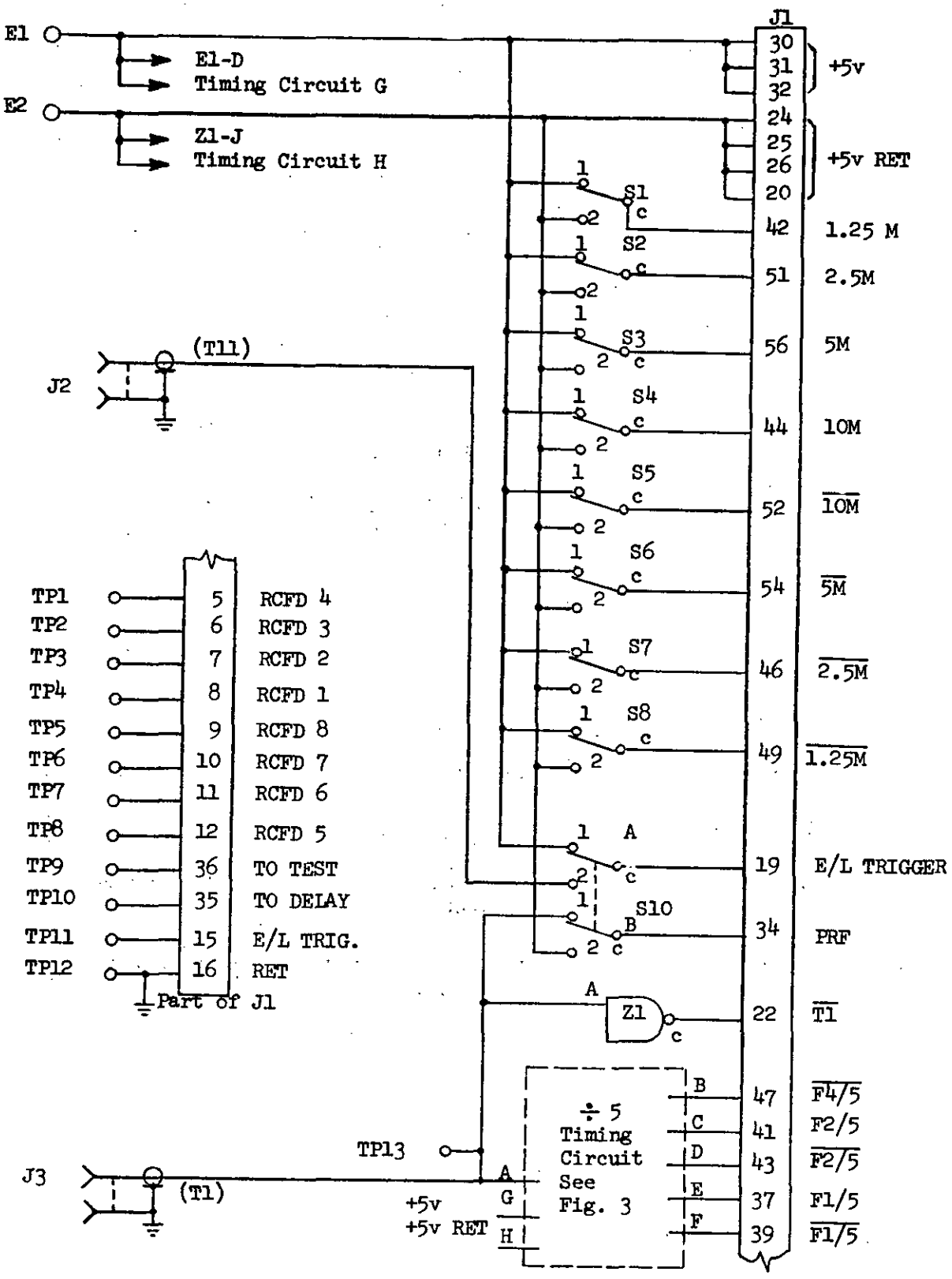


FIGURE 2. TEST FIXTURE SETUP.

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TABLE III. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1 Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
S1 thru 9	Switch SPDT.
S10	Switch DPDT.
TP1 thru TP12	Tip jack, H. H. Smith 1501 or equivalent.
Z1	Integrated circuit, SG-140-03 Sylvania or equivalent.
Z2 thru Z3	Integrated circuit, SF-50-28 Sylvania or equivalent.

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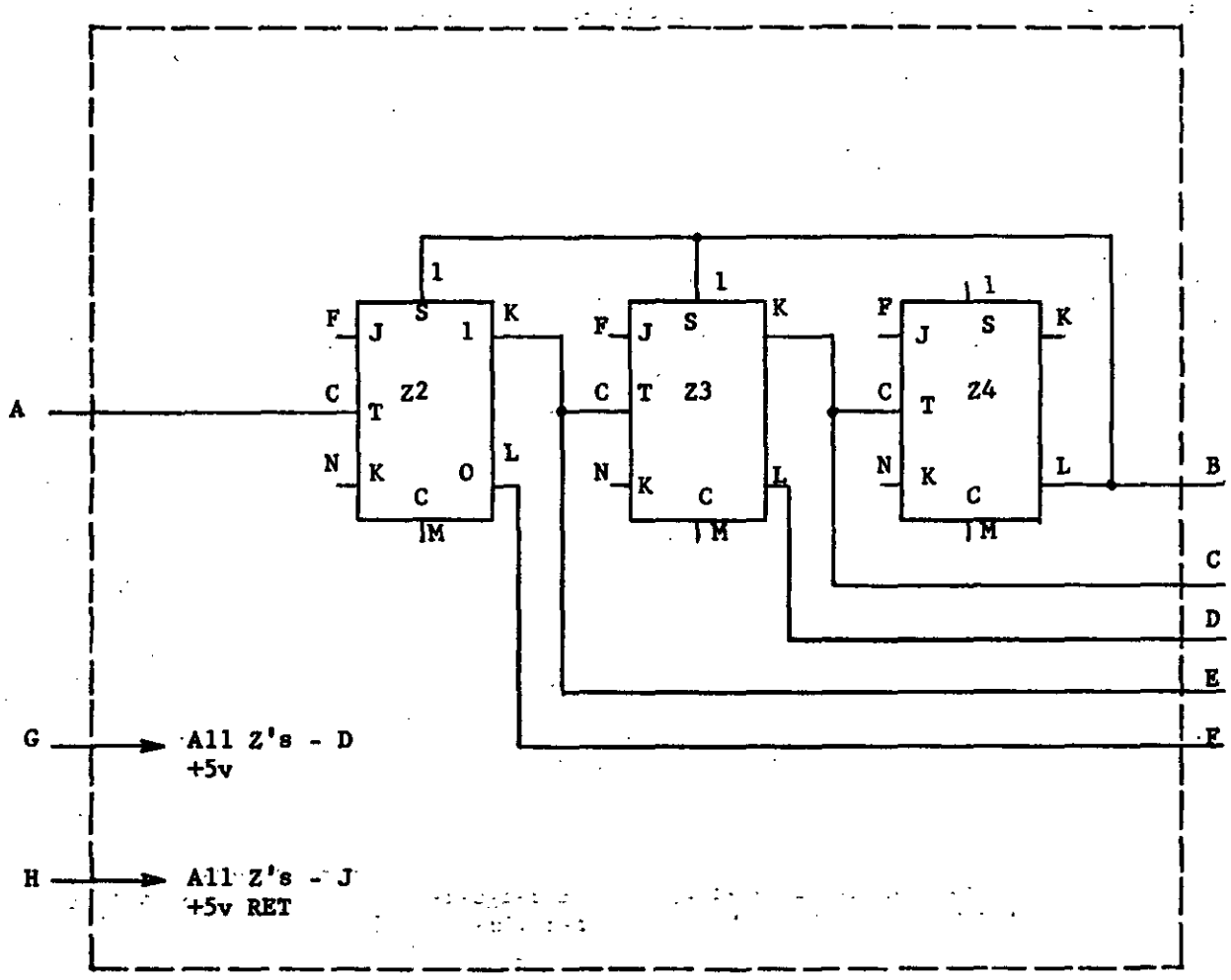


FIGURE 3. ÷ 5 TIMING CIRCUIT.

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4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected fine/fine delay assembly shall be tested as follows: Set up equipment as shown in Figure 4. Use oscilloscope, item 3, for observation of all input and output signals as specified in 3.5.2 through 3.5.6.2 except for 3.5.6.1. For observation of output signals specified in 3.5.6.1 use oscilloscope, item 4. When specified, a logic 0 shall be observed as a 0.0 ±0.5 Vdc level and a logic 1 as a +3.5 ±1.0 Vdc level. Switches S1 through S9 on the test fixture, item 5, are used to supply the logic level input signals specified in 3.5.3 through 3.5.6.2. These switches supply a logic 1 in position 1 and a logic 0 in position 2. Switch S10 in position 1 is used to apply a logic 1 level input to P1-19 and signal T1 to P1-34 for 3.5.3 through 3.5.5. In position 2, S10 applies a logic 0 level to P1-34 and signal T11 to P1-19 for 3.5.6 through 3.5.6.2.

4.6.3.1 Input power. - Using item 1, apply input power as specified in 3.5.1.

4.6.3.2 Input signals. - Using item 2, set up and apply input signals T11 and T1 as shown in Figure 1 to J2 and J3 respectively on test fixture, item 5. Verify that output signals from test fixture timing circuit for input signals $\overline{T1}$, F1/5, $\overline{F1/5}$, F2/5, $\overline{F2/5}$ and $\overline{F4/5}$ are as shown in Figure 1.

4.6.3.3 RCFD decoder output. - Using switches S1 through S8, apply logic level inputs as specified in Table 1. At TP1 through TP8 observe that output signals for terminals P1-5 through P1-12 are as specified in Table I.

4.6.3.4 T_o 2.0 μs delay output. - Use oscilloscope channel A to observe input signal at TP13 (signal T1 at P1-34) and channel B to observe output signal T_o delayed at TP9 (P1-36). Adjust R11 on the fine/fine delay assembly to obtain an output pulse as specified in 3.5.4.

4.6.3.5 T_o 0.2 μs delay output. - Using channel A and B of oscilloscope observe output pulse at TP10 and TP9 is as specified in 3.5.5.

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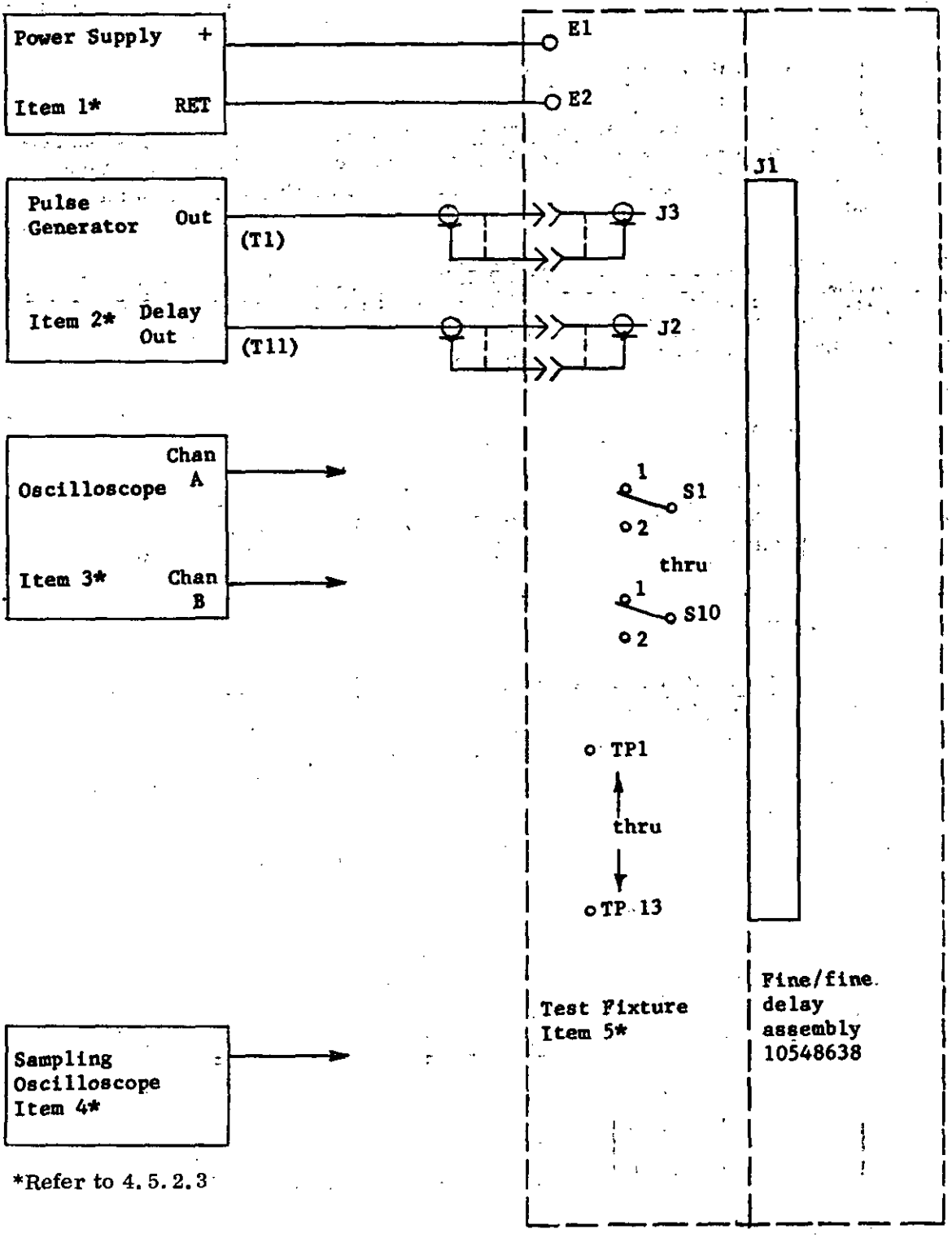


FIGURE 4. TEST SETUP.

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4.6.3.6 E/L trigger output.

4.6.3.6.1 Fine/fine delay steps. - Use oscilloscope, item 4, to observe the E/L trigger output signal at TP11 for compliance with Figure 1. Adjust capacitors C1 through C5 on the fine/fine delay assembly to position the five leading edges, from left to right respectively, to the limits shown in Figure 1.

4.6.3.6.2 E/L trigger pulse parameters. - Use oscilloscope, item 3, to observe the E/L trigger output signal, at TP-11 is as specified in 3.5.6.2. Use switch S9 in test fixture to apply logic levels to P1-3 and observe delays of output signal as specified in 3.5.6.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548638. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The fine/fine delay assembly is used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The fine/fine delay assembly supplies the first decoding operation of the range register bits, fine/fine delay of the E/L (early/late) trigger pulse during a search mode of operation of the radar and a 2 μs to delay signal.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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